

PU YANG

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About me

I am an Embodied Intelligence Algorithm Researcher at AgiBot's Embodied Intelligence Research Center. I received my Ph.D. in Computational Mathematics from the School of Mathematical Sciences, Peking University, advised by Prof. Bin Dong. My research interests include embodied intelligence and robot learning, large language models and synthetic data, reinforcement learning, and inverse problems.

Education

Peking University

Ph.D. in Computational Mathematics, advised by Prof. Bin Dong

Beijing, China

Sep. 2021 – Jul. 2026

Peking University

B.S. in Mathematics

Beijing, China

Sep. 2017 – Jul. 2021

Selected Publications

τ_0 -VLA: a Hierarchical Robot Foundation Model with World-Model-Guided Test-Time Computation

with Agibot Finch Team

- Preprint, 2026
- We present a hierarchical robot foundation model that combines world-model-guided test-time computation with a generalist low-level policy, enabling long-horizon manipulation across multiple robot embodiments.

τ_0 -WM: A Unified Video-Action World Model for Robotic Manipulation

with Agibot Finch Team

- Preprint, 2026
- We present a unified video-action world model for robotic manipulation, aiming to support scalable embodied data generation and policy learning.

Learning while Deploying: Fleet-Scale Reinforcement Learning for Generalist Robot Policies

with Agibot Finch Team

- Preprint, 2026
- We study fleet-scale offline-to-online reinforcement learning for continually improving generalist robot policies from real-world deployment experience.

Spend Wisely: Maximizing Post-Training Gains in Iterative Synthetic Data Bootstrapping

with Yunzhen Feng, Ciyuan Chen, Yuhang Wu, and Zhuoyuan Li

- Published in Neural Information Processing Systems (NeurIPS 2025, Spotlight)
- We theoretically and empirically analyze how the total generation and training budget should be allocated across iterations to maximize final performance, and demonstrate that an exponential growth policy exhibits significant advantages.

Beyond Model Collapse: Scaling Up with Synthesized Data Requires Reinforcement

with Yunzhen Feng, Elvis Dohmatob, Francois Charton, and Julia Kempe

- Published in International Conference on Learning Representations (ICLR 2025)
- We theoretically and empirically explore how feedback-augmented synthesized data can mitigate model collapse in fine-tuning large language models.

L2SR: Learning to Sample and Reconstruct for Accelerated MRI via Reinforcement Learning

with Bin Dong

- Published in Inverse Problems, 2024
- We propose an alternating training framework for jointly learning a good pair of samplers and reconstructors via deep reinforcement learning for accelerated MRI.

Work experience

Embodied Intelligence Algorithm Researcher

AgiBot Finch

2026 Spring to Present

Shanghai, China

- I am working on embodied intelligence, robot learning, and foundation models for robotic manipulation.

AI Algorithm Internship

HiSilicon

2025 Spring to Summer

Shanghai, China

- I was involved in forward-looking research on next-generation LLM architectures. My work focused on improving the accuracy and inference speed of long-context language models, including diffusion language models and sparse attention mechanisms.

AI Algorithm Internship

ByteDance

2021 Spring

Beijing, China

- I was primarily responsible for designing algorithms for audio fingerprinting, including fingerprint feature extraction, speech-library search, and parallel acceleration.

Skills

Languages: Python (PyTorch), C/C++, MATLAB, LaTeX

Human Languages: Chinese, English

Development Tools: Linux, Git

Numerical and Mathematical Skills: numerical methods, optimization, partial differential equations, probability theory

Research Areas: embodied intelligence, robot learning, reinforcement learning, large language models, synthetic data, inverse problems

Awards

Modeling competition

in the undergraduate program

- MCM, Honorable Award

Olympiad in Informatics (OI)

in high school

- NOIP, First Prize

Mathematical Competition

in high school

- China Mathematics Competition, First Prize
- China Mathematics Olympiad, Silver Medal